Charge-Dependent Sorption Controls the Bioavailability and Biotransformation of PFAS Precursors in AFFF-Impacted Solids

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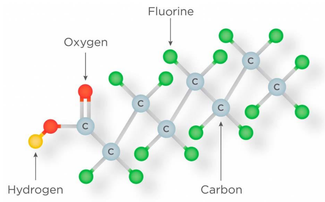
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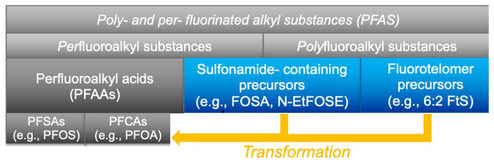
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Background

Per- and Polyfluoroalkyl Substances (PFAS)

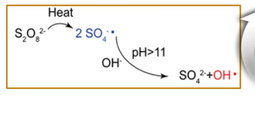
- PFAS are often called "forever chemicals" due to their extreme persistence in the environment
- Uses of PFAS: firefighting activities, water- and oil-resistant fabrics, and semiconductor manufacturing
- Toxicity:
  - Acute toxicity to aquatic species (Jones et al., 2022)
  - Increased cancer risk (Fenton et al., 2021)
  - Reduced immune system function (Kielsen et al., 2015)

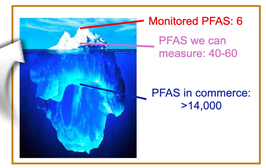




- More than half of currently used PFAS are precursors, largely due to the phaseout of terminal PFAAs (Adamson et al. 2020)
- Precursors themselves can also exhibit toxicity (Zhang et al., 2021)
- Diverse charge distributions → complicate prediction of their fate

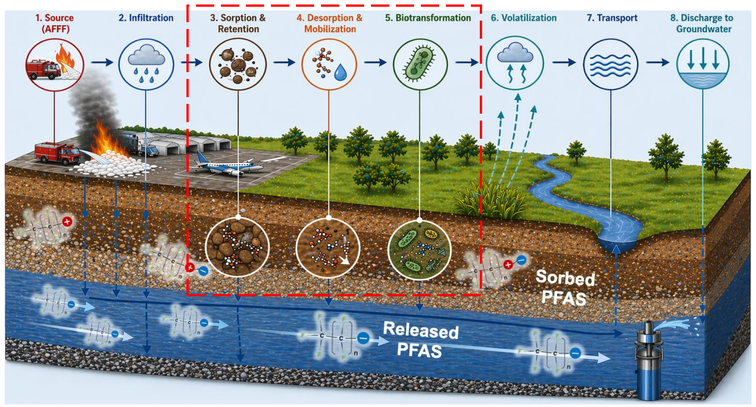
* TOP Assay: The total oxidizable precursor (TOP) assay chemically oxidizes PFAS precursors into measurable PFAAs to estimate total precursor concentrations





Objectives

- Understanding sorption/desorption and biotransformation in subsurface zones is critical for PFAS management and prediction at contaminated sites
  - *Hypothesis 1*: Do organic matter and iron minerals influence PFAS sorption?
  - *Hypothesis 2*: Do iron-reducing conditions alter PFAS sorption behavior?
  - *Hypothesis 3*: Do differences in sorption behavior lead to distinct PFAS biotransformation patterns?

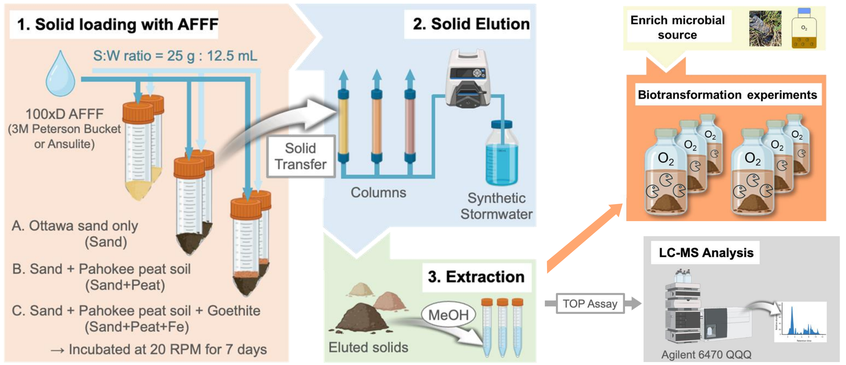


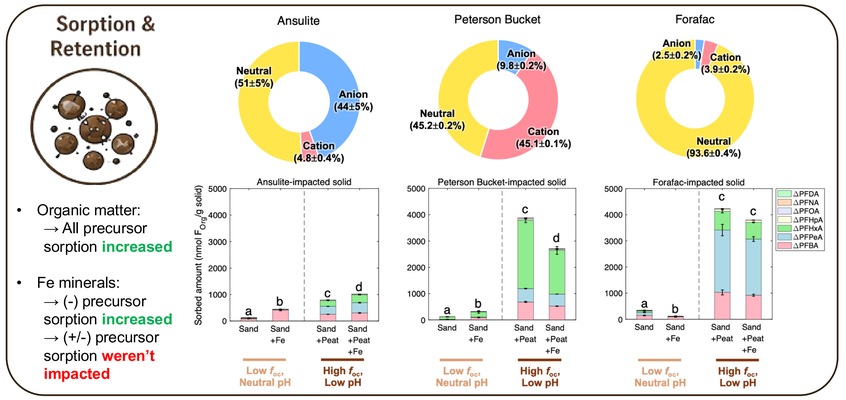
Publications

1. **Hyun Yoon**, Fuhar Dixit, Jonathan J. Uhler, Anna-Ricarda Schittich, Edmund H. Antell, Lisa Alvarez-Cohen and David L. Sedlak. 2025. Effects of organic matter, iron (hydr)oxides, and iron reductive dissolution on PFAS sorption in AFFF-impacted solids. ***Environmental Science & Technology Letters***, 13 (1), 130-136.

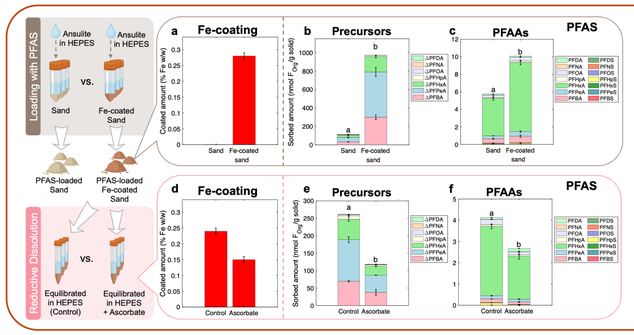
2. **Hyun Yoon**†, Anna-Ricarda Schittich†, Leyan Peng, Fuhar Dixit, Tae-Kyoung Kim, Fanshu Geng, Damian Helbling, Christopher Olivares, David L. Sedlak, and Lisa Alvarez-Cohen. Charge-Dependent Sorption Controls the Bioavailability and Biotransformation of PFAS Precursors in AFFF-Impacted Solids. *In prep.* (†Equal first author contribution)

Methodology & Results





- Organic matter: → All precursor sorption **increased**
- Fe minerals: → (-) precursor sorption **increased** → (+/-) precursor sorption **weren't impacted**



Desorption & Mobilization

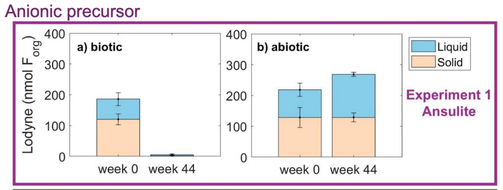
- Fe-coated sand: → (-) precursor sorption **increased**
- Fe reducing condition: → **Release** of sorbed precursor

Biotransformation

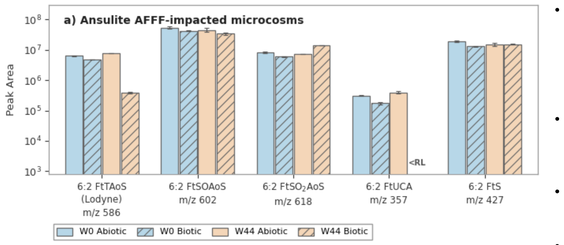
44-week aerobic microcosm experiments

- Two microcosm experiments were performed using solids pre-sorbed with AFFFs enriched in either anionic or zwitterionic precursors
- The anionic precursor Lodyne, which exhibited weaker sorption, showed strong biotransformation potential (>98% decrease)
- In contrast, the zwitterionic precursor AmPr-FHxSA exhibited limited biotransformation potential (<3% decrease), consistent with its stronger sorption behavior

Anionic precursor



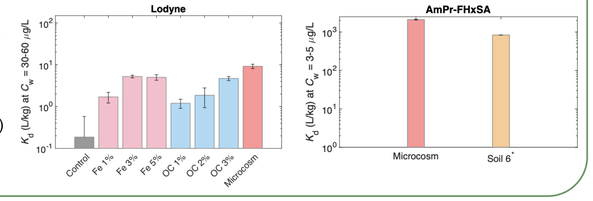
Zwitterionic precursor



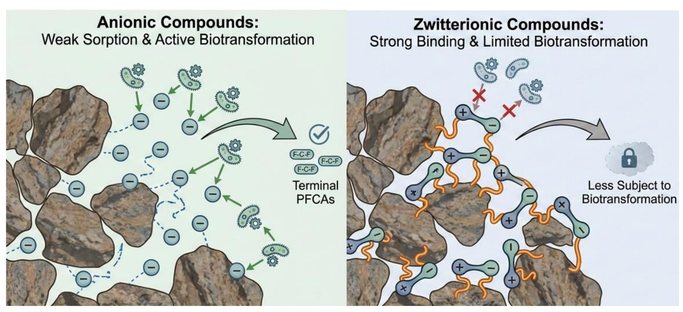
<RL = below blank-filtered reporting threshold; error bars = ±1 SD (n = 3); enrichment culture samples are single meas

- PFAAs did not increase despite substantial Lodyne depletion, prompting suspect screening analysis using HR-MS to track intermediate transformation products
- Lodyne decreased in biotic vials consistent with data from targeted analysis (stacked barplot above).
- Sulfone-intermediate (6:2 FtSO2AoS) acculmulated by a factor of 2.3
- No detectable 6:2 FtUCA remained, suggesting active biotransformation beyond the sulfone intermediate step.
- The transformation behavior observed in soil microcosms (panel a) differed substantially from that in incubations containing the same microbial enrichment amended with pure Lodyne in the absence of solids (panel b).

- K_d sorption experiments were performed to identify threshold K_d values associated with high or low biotransformation potential.
- Lodyne: K_d < 10 L/kg (high biotransformation potential)
- AmPr-FHxSA: K_d > 10^3 L/kg (low biotransformation potential)



Conclusion



- Solid significantly impacts PFAS sorption:
  - OM enhanced all precursor sorption (hydrophobic interactions)
  - Fe mineral exhibited diverse impacts (electrostatic interactions); negatively charged PFAS sorption increased while positively charged PFAS sorption decreased/unaffected
- Iron reductive dissolution led to release of sorbed PFAS
- Biotransformation pattern diverges depending on charge:
  - Lodyne (K_d < 10 L/kg): high transformation (>97%)
  - AmPr-FHxSA (K_d > 10^3 L/kg): negligible (<2%)

→ Soil chemistry-induced strong sorption can limit microbial access and transformation potential

Future work

- Collect environmental samples from contaminated sites (e.g., airports) and perform sorption experiments under varying soil and water chemistries (e.g., pH, salinity)
- Conduct biotransformation experiments using pre-grown microbial enrichments to evaluate precursor transformation potential under different pH conditions

